

Proof. Statement. Let

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n$$

be real numbers. We must show

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right). \quad (1)$$

1. Trivial and degenerate cases.

- If $n = 1$, (1) reduces to $a_1 b_1 \geq a_1 b_1$, true with equality.
- If all a_i are equal, say $a_i = a$, then $\sum a_i b_i = a \sum b_i$ and both sides of (1) equal $na \sum b_i$.
- The case “all b_i equal” is symmetric.

Hence we may assume $n \geq 2$ and that neither sequence is constant.

2. Rearrangement Inequality. For increasing sequences $x_1 \leq \cdots \leq x_n$ and $y_1 \leq \cdots \leq y_n$ and any permutation $\sigma \in S_n$,

$$\sum_{i=1}^n x_i y_{\sigma(i)} \leq \sum_{i=1}^n x_i y_i \leq \sum_{i=1}^n x_i y_{n+1-i}.$$

Thus the sum with the identity permutation is maximal. Applying this with $x_i = a_i$ and $y_i = b_i$ yields

$$\sum_{i=1}^n a_i b_i \text{ is the largest of the } n! \text{ numbers } \sum_{i=1}^n a_i b_{\sigma(i)} \text{ } (\sigma \in S_n). \quad (2)$$

3. Average over all permutations. Consider

$$S := \sum_{\sigma \in S_n} \sum_{i=1}^n a_i b_{\sigma(i)}. \quad (3)$$

Fix an index i . In the collection of all permutations each b_j appears in position i exactly $(n-1)!$ times, hence

$$\sum_{\sigma \in S_n} b_{\sigma(i)} = (n-1)! \sum_{j=1}^n b_j.$$

Multiplying by a_i and summing over i gives

$$S = (n-1)! \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right). \quad (4)$$

Dividing by the number of permutations $n!$ we obtain the average value:

$$\frac{S}{n!} = \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right). \quad (5)$$

4. Comparison with the maximal sum. From (2) the ordered sum $\sum a_i b_i$ is not smaller than any other $\sum a_i b_{\sigma(i)}$; therefore it is not smaller than their average. Using (5),

$$\sum_{i=1}^n a_i b_i \geq \frac{S}{n!} = \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right).$$

Multiplying by n yields exactly inequality (1).

5. Equality condition. Equality in the Rearrangement Inequality (2) occurs iff the two sequences are not strictly ordered with respect to each other, i.e. at least one of the sequences is constant. Consequently equality in Chebyshev's inequality (1) holds precisely when all a_i are equal or all b_i are equal – the situations already treated in Step1. No other case can give equality, because then the ordered sum would be strictly larger than the average in Step4.

6. Conclusion. Using the Rearrangement Inequality together with a simple counting argument we have proved that for any two similarly ordered real sequences

$$a_1 \leq a_2 \leq \dots \leq a_n, \quad b_1 \leq b_2 \leq \dots \leq b_n,$$

the Chebyshev sum inequality

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right)$$

holds, with equality exactly when one of the sequences is constant. $\square$

$\square$